3. Dik : Himpunan pelanggan

- $A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\} \rightarrow A = \{29, 31, 33, 36, 39, 42, 44\}$
- $B = \{N+3, N+5, N+7, N+10, N+12, N+18\} \rightarrow B = \{32, 34, 36, 39, 41, 47\}$
- $C = \{N+1, N+7, N+10, N+11, N+17, N+20\} \rightarrow C = \{30, 36, 39, 40, 46, 49\}$
- Semesta $= \{N, N+1, \ldots N+20\}$

a. Dit : gambar diagram venn, $|A \cup B \cup C|$, $|A \cap B \cap C|$, $|(A \cup B) \cap C'|$, $|A \oplus B|$

Dij :

- Mencari irisan himpunan
  - $A \cap B$,

    $A = N, N+2, N+4, N+7, N+10, N+13, N+15$

    $B = N+3, N+5, N+7, N+10, N+12, N+18$
  - $A \cap B = \{N+7, N+10\} \rightarrow \{36, 39\}$
  - $A \cap C = \{N+7, N+10\} \rightarrow \{36, 39\}$
  - $B \cap C = \{N+7, N+10\} \rightarrow \{36, 39\}$
  - $A \cap B \cap C = \{N+7, N+10\} \rightarrow \{36, 39\}$

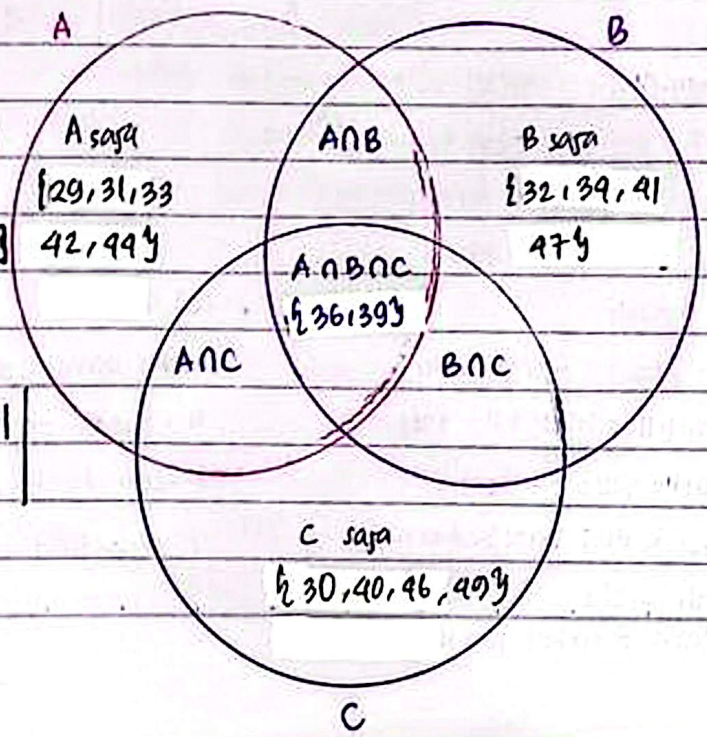
    $|A \cap B \cap C| = 2$
- Menghitung $A \cup B \cup C$

  Gabungkan semua elemen

  $A \cup B \cup C = \{N, N+1, N+2, N+3, N+4, N+5, N+7, N+10, N+11, N+12, N+13, N+15, N+17, N+18, N+20\}$

  $A \cup B \cup C = \{29, 30, 31, 32, 33, 34, 36, 39, 40, 41, 42, 44, 46, 47, 49\} \rightarrow |A \cup B \cup C| = 15$
- Menghitung $(A \cup B) \cap C'$
  - $A \cup B = \{29, 31, 32, 33, 34, 36, 39, 41, 42, 44, 47\}$
  - $C' = U - C$
  - $C' = \{29, 31, 32, 33, 34, 35, 37, 38, 41, 42, 43, 44, 45, 47, 48\}$
  - $(A \cup B) \cap C' = \{29, 31, 32, 33, 34, 41, 42, 44, 47\}$
  - $|(A \cup B) \cap C'| = 9$
- Menghitung $A \oplus B$
  - $A \oplus B = (A \cup B) - (A \cap B)$
  - $|A \oplus B| = 11 - 2 = 9$

$A \oplus B = \{29, 31, 32, 33, 34, 41, 42, 44, 47\}$



Program Venn Tiga Himpunan

Ringkasan Hasil

1. $\lvert A \cup B \cup C \rvert$	2. $\lvert A \cap B \cap C \rvert$	3. $\lvert (A \cup B) \cap C' \rvert$
$= 15$	$= 2$	$= 9$

4. $|A \oplus B|$

$= 9$

SiDU

c. Dik : $f(x) = kx + (a+b)$, $g(x) = \frac{x^2}{b+1} + a$

- $a = 2$, $b = 9$, $K = 12$
- $f(x) = kx + 11$, $g(x) = \frac{x^2}{10} + 2$

Dit : $(f \circ g)(x)$, $(g \circ f)(x)$, apakah komutatif ? (uji dengan $x = b+2 = 11$)

Dij : - Komposisi $(f \circ g)(x) = f(g(x))$

$f(g(x)) = 12\left(\frac{x^2}{10} + 2\right) + 11$

$(f \circ g)(x) = \frac{6}{5}x^2 + 35$

- Komposisi $(g \circ f)(x) = g(f(x))$

$g(f(x)) = \frac{(12x + 11)^2}{10} + 2$

Hasil Akhir

$(g \circ f)(x) = \frac{144x^2 + 264x + 121}{10} + 2$

$= \frac{144x^2 + 264x + 141}{10}$

- Uji komutatif $(x = 11)$

- Hitung $(f g(11)) = \frac{6}{5}(121) + 35$

$= 145.2 + 35 = 180.2$

- Hitung $g(f(11)) = \frac{144(121) + 264(11) + 141}{10}$

$= \frac{20493}{10}$

$= 2049.3$

- Kesimpulan

$(f \circ g)(x) \neq (g \circ f)(x) \rightarrow 180.2 \neq 2049.3$

Hasilnya berbeda. Tidak komutatif

d. Dik : $f(x) = kx + (a+b) \rightarrow f(x) = 12x + 11$

Dit : Tentukan $f^{-1}(x)$, jika pendapatan $y = K \cdot 100 + a + b$ berapa nilai x ?

Dij : - Mencari inverse fungsi $f^{-1}(x)$

$y = 12x + 11$

Cari x

$y - 11 = 12x$

$x = \frac{y - 11}{12}$

- Pendapatan

$K(100) + a + b$

$12(100) + 2 + 9$	Cari volume pembelian	Jika pendapatan pelanggan
1211	$12x + 11 = 1211$	$= K \cdot 100 + (a+b)$ artinya pelanggan
	$12x = 1200$	membeli 100 unit. Karena fungsi linear
	$x = 100$	(harga per unit tetap)

e. Dik : $f(x) = 12x + 11$

$g(x) = \frac{x^2}{10} + 2$

domain dan kodomain : $R \to R$

Dit : Injektif, surjektif, dan bijektif

Dis : 1) Klasifikasikan fungsi $f(x) = 12x + 11$

(a) Injektif : $f(x_1) = f(x_2)$

harus terbukti $x_1 = x_2$

$f(x_1) = f(x_2)$

$12x_1 + 11 = 12x_2 + 11$

$12x_1 = 12x_2$ (dibagi 12)

$x_1 = x_2$

f injektif

(b) Surjektif

Untuk setiap $y \in R$ harus ada $x \in R$

sehingga $f(x) = y$

$y = 12x + 11$

$x = \frac{y-11}{12}$

karena $x \in R$ untuk setiap

$y \in R$ maka kodomain punya

Pra-citra.

f surjektif

(c). Bijektif

karena injektif dan surjektif,

f bijektif

2) Klasifikasi fungsi

$g(x) = \frac{x^2}{10} + 2$

(a) Injektif

$x_1 = 2$, $x_2 = -2$

$g(2) = \frac{4}{10} + 2 = 2.4$

$g(-2) = \frac{4}{10} + 2 = 2.4$

$g(2) = g(-2)$

$2 \neq -2$ (melanggar definisi injektif)

Maka g tidak injektif

(b) Surjektif

$x^2 \geq 0$

$\frac{x^2}{10} \geq 0$

$g(x) \geq 2$

Rang fungsi $[2, \infty)$

kodomain yg diberikan R

maka nilainya 0, -3, 1

karena tidak semua elemen

kodomain tercapai, jadi g tidak

surjektif

(c) Bijektif

Bijektif perlu injektif dan surjektif

karena keduanya gagal

g tidak bijektif

SiDU